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-----|turtlebot3-msgs
```

```
-----|turtlebot3 messages ....
```

Running ROS Master + ROS packages (<http://robolab.cse.unsw.edu.au:4443/comp3431-rsa/comp3431-rsa/wikis/website/ros#running-ros-master-ros-packages>)

Before you can call any ROS commands/software, such as `roscore`, `roslaunch` or even `catkin_make`, you need to source the correct ROS setup script. If you use the above build configuration, then you should call `source ~/workspace/devel/setup.bash` when you start each terminal. You can automate this by adding the source like to your `.bashrc`.

ROS needs two environment variables to be configured, `ROS_IP` and `ROS_MASTER_URI`. `ROS_IP` tells ROS the IP of your current machine. You can find this out with the `ifconfig` command. `ROS_MASTER_URI` tells the ROS process running in the current terminal where the ROS Master is located. Configure these with environment variables with:

```
ifconfig
export ROS_IP=<ip-machine>
export ROS_MASTER_URI=http://<ip-ros-master>:11311
```

For ROS to function it needs a common ROS Master (ie. a `roscore`). If no master is running, then a ROS Master is started. We recommend starting **one and only one** ROS master is a separate terminal on your computer, and then connect every other ROS process to that master. That is, open a terminal and run: `roscore`

TL;DR

When you start a terminal you should run:

```
source ~/workspace/devel/setup.bash
export ROS_IP=<ip-machine>
export ROS_MASTER_URI=http://<ip-ros-master>:11311
```

Start a ROS master in a separate terminal on your computer, and use new terminals for all other ROS processes.

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